OD Inverse Estimation Model Benchmark

Technical report connecting the underdetermined OD↔turning inverse problem, the implementations in `src/ml/rigorous_benchmark/`, and the measured artifacts in `outputs/benchmark/`. Figures live in `docs/figures/od_benchmark/` and are regenerated by `scripts/generate_od_benchmark_doc_figures.py`.

Artifact root	`outputs/benchmark/`
Seed / split	seed 42; 70 / 15 / 15 on 100 000 FP synthetics
Survey	turning index 100000; held out until inference
Forward map	$\hat{\mathbf{x}} = \hat{\mathbf{Y}}_{\text{flat}}\, A^{\top}$ (turning space)

Ranking stance (artifacts only — no single "best" model):

- Synthetic test OD MAE: `mlp` best (646.575605); AE / PLS worst.
- Synthetic test composite: `physics_ridge` ≈ `ridge` best (∼1.506); `mlp` third.
- Synthetic forward: `moore_penrose` / `physics_ridge` / `ridge` ≈ 0.
- Survey OD MAE: `nullspace_mlp` best (395.114834); `ridge` / `physics_ridge` worst (∼664).
- Survey forward: operator / ridge family ≈ 0; PLS / AE poor.

All numeric claims are from CSV/JSON/NPY artifacts or source code. Missing fields are marked **Not available in the current experiment artifacts.**

Problem statement

We estimate a non-negative origin–destination (OD) matrix $\mathbf{Y} \in \mathbb{R}^{n\times n}$ ($n = 24$ Sioux Falls zones; flattened $\mathbf{y} \in \mathbb{R}^{576}$) from observed turning counts $\mathbf{x} \in \mathbb{R}^{m}$ ($m = 178$) under a fixed linear assignment operator $A \in \mathbb{R}^{m\times 576}$:

$$\mathbf{x} = A\mathbf{y} \quad \text{(exact for clean synthetics).}$$

The map $\mathbf{y} \mapsto A\mathbf{y}$ is **underdetermined**: $\text{rank}(A) < 576$, so infinitely many OD matrices are consistent with the same turns. Estimation therefore requires a prior (supervised learning on synthetics, Tikhonov penalty, null-space parameterization, etc.).

Symbol	Meaning
$\mathbf{x}$	Turning-count vector (observation)
$\mathbf{y}$, $\mathbf{Y}$	Flattened / matrix OD demand
A, A_{turn}	Fractional assignment operator
A^{+}	Moore–Penrose pseudoinverse
$\hat{\mathbf{y}}$, $\hat{\mathbf{Y}}$	Estimated OD
$\hat{\mathbf{x}} = \hat{\mathbf{y}}\, A^{\top}$	Forward reconstruction in **turning space**

Geometry of the inverse problem

From `outputs/benchmark/operator/operator_meta.json` (SVD of the saved A for this run):

Quantity	Value
Shape	178×576
Numerical rank (rank)	170
Nullity	406
Condition number	$9.214070180877218 \times 10^{9}$
Effective rank (99% energy)	105

Prior methodology documents on this network report fractional-assignment analysis rank ≈ 174 and nullity ≈ 402 (`docs/synthetic_od_method.tex`, `docs/od_estimation_methodology.tex`). Those are analysis/log references, not this benchmark's SVD dump; this document privileges `operator_meta.json` and notes the historical 174 for continuity.

The general solution of $A\mathbf{y} = \mathbf{x}$ is

$$\mathbf{y} = A^{+}\mathbf{x} + N\mathbf{z}, \qquad N = \text{null}(A),\ \mathbf{z}\text{ free},$$

which is exactly the parameterization used by `nullspace_mlp`.

```
flowchart LR
  Y["OD y in R^576"] -->|A| X["Turns x in R^178"]
  X -->|A+ particular| Yp["y_part = A+ x"]
  Yp -->|+ N z| Yhat["ŷ = y_part + N z"]
```

Experimental protocol

```
flowchart TB
  FP["FP synthetics 100k"] --> Split["Random split seed 42\n70k / 15k / 15k"]
  Split --> HPO["HPO on train subsample + val"]
  HPO --> Fit["Final fit trainuval\n(capped)"]
  Fit --> Test["One-shot synthetic test"]
  Fit --> Survey["Survey inference\nindex 100000"]
  A["A_turn fixed"] --> Fit
  A --> Test
  A --> Survey
```

Facts from `data/leakage_report.json` and `run_config.json`:

- FP synthetics only for train/val/test/HPO; survey excluded (`survey_in_train_val_test: false`).
- Scalers fit on **train only**; never refit on survey.
- Composite selection on validation only; test never used for HPO.
- OOM-safe caps: `n_trials=8`, `hpo_train_subsample=5000`, `hpo_val_cap=3000`, `final_train_cap=25000`.
- Full-run wall time (`run_summary.json`): 628.862 s for 13 models.

Data: first-principles OD and fractional assignment

First-principles OD generation

Implementation: `src/data/first_principles/` + `src/data/generate_fp_od_dataset.py`.

Stage	Module	Role
A	`latent.py`	Latent production / attraction factors
B	`gravity.py`	Gravity kernel + reciprocity
C	`heavy_tails.py`	Gamma corridor weights
D	`compositional.py`	Dirichlet compositional noise
E	`generator.py`	Scaling + IPF
F	CLI filters	Dataset quality filters

Training artifact: `outputs/od_generator_fp/synthetic_od_fp_synthetics_only.npy`, shape $(100000, 24, 24)$.

Turning operator A

Built by `src/data/generate_turning_counts.py` / `src/data/fractional_assignment.py` (k-shortest paths + logit route choice → fractional A_{turn}).

Paths from `run_config.json`:

- `A_turn`: `outputs/turning_counts_fp/A_turn.npy`
- Turning counts: `outputs/turning_counts_fp/turning_counts.npy`
- SHA-256 of A: `09b54843d68eac3904e0a55b941445de41a47d55cc1e8877e80580028e8e1eb9`

Forward sanity (`leakage_report.forward_sanity`): 32 samples checked; `allclose: true`; formula recorded as `X = Y_flat @ A_turn.T`.

Splits, leakage, and preprocessing

Split	n	Source
Train	70 000	`leakage_report.json`
Val	15 000	
Test	15 000	
Survey turning index	100 000	held out

- Indices: `outputs/benchmark/data/split_indices.npz` (`train_idx`, `val_idx`, `test_idx`, `seed`).
- Generator-aware split: **N/A — no per-sample seed metadata in FP artifacts; random split v1** (`leakage_report`).
- Standardization: `standardize_x=true`, `standardize_y=true` (`run_config.json`); scalers in `preprocessing/scalers.joblib`.
- Survey scale diagnostics (`survey_inference/scale_diagnostics.json`): survey x mean shift vs train -1127.62; std ratio 0.588; approximate Wasserstein-1 on x 1268.88. Scalers **not** refit.

Models

This section documents each estimator as implemented: motivation, equations, architecture (where applicable), training objective, constraints, and the hyperparameters selected for the reported run.

Shared neural training loss (`composite_loss` in `src/ml/neural_trainer.py`), used by MLP-family / AE / nullspace models unless noted:

$$\mathcal{L} = w_{\text{od}}\,\mathrm{MSE}(\hat{\mathbf{y}}, \mathbf{y}) + w_{\text{fwd}}\,\mathrm{MSE}(A\hat{\mathbf{y}}, \mathbf{x}) + w_{\text{prod}}\,\mathrm{MSE}(\hat{\mathbf{P}}, \mathbf{P}) + w_{\text{attr}}\,\mathrm{MSE}(\hat{\mathbf{Q}}, \mathbf{Q}),$$

with defaults $w_{\text{od}} = 1.0$, $w_{\text{prod}} = w_{\text{attr}} = 0.1$, and w_{fwd} HPO'd per model. Hidden activations in trunks are **GELU** (`NeuralTrainConfig.activation`). Output nonnegativity is controlled separately by `output_activation` ∈ {`none`, `relu`, `softplus`} after inverse scaling (and after optional residual add). Diagonal cells are zeroed; optional support mask zeros OD pairs never observed in training.

Moore–Penrose pseudoinverse

Code: `src/ml/rigorous_benchmark/models/pinv.py` → `pinv_predict`.

Role. Unsupervised particular solution of the linear inverse problem. No trainable parameters and no use of OD labels.

Estimator.

$$\hat{\mathbf{y}} = A^{+}\mathbf{x}.$$

A^{+} is precomputed once from the SVD of A and stored under `outputs/benchmark/operator/`. Among all least-squares solutions of $A\mathbf{y} = \mathbf{x}$, the Moore–Penrose solution is the unique minimum-ℓ_2-norm particular solution; the free null-space component is identically zero.

Architecture. None (closed-form linear algebra).

Constraints. `constraint_strategy=none`: negatives and diagonal leakage are allowed in the reported main run (ablations explore post-hoc clips).

HPO. None. Best params: `{}`.

What it measures. Pure operator consistency. Near-zero forward RMSE is expected; OD MAE reflects how far the FP / survey prior sits from the minimum-norm particular solution.

Tikhonov regularization

Code: `src/ml/rigorous_benchmark/models/tikhonov.py` (`tikhonov_solve` via `cho_solve`).

Role. Classical unsupervised regularized least squares on the known operator. **Does not use OD labels** at fit or predict time. Distinct from supervised Ridge (below).

Estimator.

$$\hat{\mathbf{y}} = \arg\min_{\mathbf{y}} \|A\mathbf{y} - \mathbf{x}\|_2^2 + \lambda\|\mathbf{y}\|_2^2 \quad \Leftrightarrow \quad (A^{\top}A + \lambda I)\mathbf{y} = A^{\top}\mathbf{x}.$$

Implementation factors $A^\top A + \lambda I$ once (Cholesky) and solves against right-hand sides XA for batches of **x**. Identity Tikhonov matrix $L = I$ only.

Architecture. None (closed-form linear solve).

HPO. $\lambda \in [10^{-6}, 10^{6}]$ log-uniform, selected by validation composite.
Selected: $\lambda = 1.0540913515845597 \times 10^{-6}$ (`tikhonov/best_params.json`).
Separate `hpo_best.json` best_value: **Not available in the current experiment artifacts.**

What it measures. Same geometry as pinv when $\lambda \to 0$; small λ here yields near-pinv behaviour with mild damping of high-frequency singular directions.

Supervised Ridge

Code: `src/ml/rigorous_benchmark/models/ridge.py` (sklearn multi-output Ridge).

Role. Learn a linear map from turning counts to OD from labelled synthetics. A **is not used at predict time** — physics enters only indirectly through the synthetic (X, Y) pairs.

Estimator. Fit on scaled **x** → raw **y**:

$$\hat{\mathbf{y}} = W\mathbf{x}_{\text{scaled}} + \mathbf{b}, \qquad \min_{W,\mathbf{b}} \|\mathbf{Y} - XW^\top\|_F^2 + \alpha\|W\|_F^2$$

(sklearn's multi-target Ridge; each OD cell is a separate target sharing the same α).

Architecture. Linear multi-output regressor: input dim 178 → output dim 576 (plus intercept). No hidden layers.

HPO. $\alpha \in [10^{-4}, 10^{5}]$ log-uniform.
Selected: $\alpha = 0.00010403002134459767$.

Constraints. none in the main run (negatives allowed). Ablations apply relu/clip/IPF post-hoc.

What it measures. How well a linear supervised prior matches the FP distribution and how much that prior conflicts with survey OD under distribution shift.

Physics Ridge

Code: `src/ml/rigorous_benchmark/models/physics_ridge.py`.

Role. Hybrid: supervised Ridge prior $\mathbf{y}_0$, then a **closed-form** refine that pulls the prediction toward the forward constraint $A\mathbf{y} \approx \mathbf{x}$.

Two-stage estimator.

1. Fit Ridge as above → $\mathbf{y}_0$.
2. Refine:

$$\mathbf{y} \leftarrow \arg\min_{\mathbf{y}} \|\mathbf{y} - \mathbf{y}_0\|_2^2 + \mu\|A\mathbf{y} - \mathbf{x}\|_2^2 \quad \Leftrightarrow \quad (I + \mu A^\top A)\mathbf{y} = \mathbf{y}_0 + \mu A^\top \mathbf{x}.$$

Critical distinction from Tikhonov. Tikhonov has no supervised $\mathbf{y}_0$ and shrinks toward **0**. Physics Ridge shrinks toward the Ridge prediction while enforcing forward consistency with strength μ.

Architecture. Same linear Ridge map as above, plus one Cholesky solve per prediction batch (no neural net).

HPO. $\alpha \in [10^{-4}, 10^{5}]$, $\mu \in [10^{-6}, 10^{2}]$ (log).
Selected: $\alpha = 0.00011692997958212092$, $□=50.62467440787875‘.

What it measures. Whether a strong forward refine can keep composite / forward near perfect while retaining Ridge's synthetic OD accuracy. Survey results show that refine does **not** rescue a mismatched prior.

Partial least squares (PLS)

Code: `src/ml/rigorous_benchmark/models/pls.py` (`sklearn.cross_decomposition.PLSRegression`).

Role. Supervised latent linear map $X \to Y$ with a small number of latent components; useful when X and Y are jointly collinear.

Estimator. PLS with `scale=False`, `n_components` HPO'd. Predict: scaled **x** → raw **y**. No use of A at predict.

Architecture (conceptual).

```
x (178) ──► [PLS latent space, k components] ──► y (576)
```

Not a neural net; sklearn's NIPALS-style PLS.

HPO. `n_components` ∈ [2, max feasible ≤ 20].
Selected: `n_components=20`.

What it measures. Low-rank linear transfer. On this run it underfits synthetic OD/forward relative to Ridge/MLP, but transfers relatively well on the single survey matrix (second-best survey OD MAE).

Direct MLP

Code: `src/ml/rigorous_benchmark/models/mlp.py` + `DirectODRegressor` in `src/ml/neural_core.py` with `residual_blocks=False`.

Role. Nonlinear supervised map **x** $\mapsto$ **y** that can place mass in the null space of A according to the FP prior. A enters **only through the forward term of the loss**, not as a hard constraint at predict.

Architecture (selected run).

```
Input: scaled turning counts  x ∈ R^{178}
        |
        ▼
   Linear 178 → 512 + GELU
        |
        ▼
   Linear 512 → 1024 + GELU
        |
        ▼
   Linear 1024 → 576          (logits in scaled-Y space)
        |
        ▼
   Inverse Y-scaler → raw OD
```

```
       |
       ▼
   output_activation = ReLU
       |
       ▼
   Zero diagonal; optional support mask
       |
       ▼
Output: ŷ ∈ R^{576}
```

Layer widths come from HPO `hidden0=512`, `hidden1=1024`. Trunk activation is GELU (config default). No residual blocks.

Training. Adam; early stopping on validation OD MAE; composite loss with HPO'd `forward_weight`. Final fit uses train∪val capped at 25 000 samples.

Selected hyperparameters (`mlp/best_params.json`):

Hyperparameter	Value
hidden0 / hidden1	512 / 1024
lr	0.0016755052359850296
weight_decay	1.1756010900231857e-05
batch_size	256
forward_weight	1.6043939615080793
output_activation	relu

Best val MAE during fit: 642.105225 (in same file). Parameter count in CSV: **Not available in the current experiment artifacts.**

What it measures. Strongest synthetic OD prior among reported models; pays a forward-error cost (~340 RMSE) because the hypothesis class is unconstrained in the null space.

Residual MLP

Code: `src/ml/rigorous_benchmark/models/residual_mlp.py` + `DirectODRegressor(..., residual_blocks=True)`.

Role. Same supervised $X \to Y$ problem as the MLP, but with **architectural** residual blocks (ResBlocks) for deeper feature transforms at fixed width. This is **not** residual learning from a base OD matrix.

Architecture (selected run: width 512, 3 ResBlocks).

Each `ResBlock(dim)` (`neural_core.py`):

$$h \leftarrow h + W_2\,\sigma(W_1 h), \qquad \sigma = \text{GELU}.$$

Full forward:

```
Input: x ∈ R^{178}
       |
       ▼
   Linear 178 → 512 + GELU          (in_proj)
       |
```

```
        ▼
   ResBlock × 3  (width 512)        # architectural skip connections
        │
        ▼
   Linear 512 → 576                 (out_proj logits)
        │
        ▼
   Inverse Y-scaler → raw OD
        │
        ▼
   softplus; zero diagonal; support mask
        │
        ▼
Output: ŷ ∈ R^{576}
```

HPO forces `hidden0 = hidden1 = width` ∈ {128, 256, 512} and `num_res_blocks` ∈ {1, 2, 3}.

Selected hyperparameters:

Hyperparameter	Value
width (= hidden0 = hidden1)	512
num_res_blocks	3
lr	0.0009616005310543457
weight_decay	4.857295179217169e-06
batch_size	128
forward_weight	1.3945464785138597
output_activation	softplus
parameters (CSV)	1 963 072

What it measures. Whether deeper residual capacity improves over the plain MLP under the same composite objective. On this run it is slightly worse than MLP on synthetic OD MAE (657 vs 647) with similar survey behaviour.

Physics Residual MLP

Code: `src/ml/rigorous_benchmark/models/physics_residual_mlp.py`.

Role. Two "physics" ingredients on top of Residual MLP:

1. **Elevated forward weight** HPO range `physics_forward_weight` ∈ [0.5, 3.0] (vs shared neural range [0, 1.5]).
2. **Residual learning from a fixed base OD:** `residual_learning=True` in `logits_to_od` adds `base_od_flat`, which `data.py` sets to the **survey** OD vector. The network therefore predicts a correction $\Delta\mathbf{y}$ around that base:

$$\hat{\mathbf{y}} = \mathrm{act}\big(\mathrm{unscale}(f_\theta(\mathbf{x})) + \mathbf{y}_{\mathrm{survey}}\big) \quad \text{(then zero diagonal / mask).}$$

Architecture (selected: width 512, 1 ResBlock). Same ResBlock trunk as Residual MLP, but shallower:

```
x (178) → Linear→512+GELU → ResBlock×1 → Linear→576
         → unscale → + y_survey → ReLU → diag0 / mask
```

Selected hyperparameters:

Hyperparameter	Value
width	512
num_res_blocks	1
lr	0.0011256123615936324
weight_decay	7.947147424653752e-05
batch_size	64
forward_weight	2.740228249808733
output_activation	relu
parameters (CSV)	912 448

Distinction from Residual MLP. Residual MLP = ResBlocks only. Physics Residual MLP = ResBlocks **+** survey-base residual add **+** higher forward weight. Anchoring to survey OD is a strong inductive bias for survey inference but couples the synthetic training objective to a single external matrix.

Autoencoder family (ae_32, ae_64, ae_128, ae_64_finetune)

Code: `src/ml/rigorous_benchmark/models/autoencoder.py` + `ODEncoder` / `ODDecoder` / `LatentPredictor` / `LatentODModel` in `neural_core.py`.

Role. Compress OD matrices into a latent bottleneck, then learn to predict that latent from turning counts. Intended to enforce a low-dimensional OD manifold prior.

Architecture. Fixed AE widths from config `latent_hidden=(256, 128)` (not HPO'd). Latent dim $d \in \{32, 64, 128\}$.

Stage 1 — OD autoencoder pretrain (labels only; no X):

```
y_scaled (576)
  → Linear 576→256 + GELU → Linear 256→128 + GELU → Linear 128→d     (encoder)
  → Linear d→128 + GELU → Linear 128→256 + GELU → Linear 256→576     (decoder)
  → logits_to_od → MSE(pred, y_raw)
```

Stage 2 — Latent predictor (decoder frozen):

```
x_scaled (178)
  → Linear 178→h0 + GELU → Linear h0→h1 + GELU → Linear h1→d      (LatentPredictor)
  → frozen decoder → logits_to_od
Loss = MSE(z_pred, z_encoder(y)) + composite_loss(ŷ, y, x, A)
```

Stage 3 — optional finetune (`ae_64_finetune` only): unfreeze decoder; train end-to-end with composite loss for `finetune_epochs` (default 15) at `finetune_lr` (default 10^{-4}).

Selected run configs (from each `best_params.json`):

Model	Latent d	Finetune	h0 / h1	forward_weight	output_act	parameters
ae_32	32	no	512 / 512	0.277282	softplus	556 000
ae_64	64	no	512 / 512	0.277282	softplus	576 512
ae_128	128	no	512 / 512	0.456921	relu	617 536

Model	Latent d	Finetune	h0 / h1	forward_weight	output_act	parameters
ae_64_finetune	64	yes	512 / 512	0.456921	relu	576 512

What it measures. Whether a low-dim OD manifold helps. Under this HPO budget, all AE variants underperform MLP/Ridge on synthetic OD and show large forward errors — bottleneck capacity and/or training protocol appear insufficient for the underdetermined inverse.

Nullspace MLP

Code: `src/ml/rigorous_benchmark/models/nullspace_mlp.py`.

Role. Explicitly parameterize the affine solution set of $A\mathbf{y} = \mathbf{x}$. The network predicts only null-space coefficients; the particular solution is hard-wired.

Estimator.

$$\hat{\mathbf{y}} = A^{+}\mathbf{x} + N\, f_\theta(\mathbf{x}_{\text{scaled}}),$$

where $N \in \mathbb{R}^{576\times k}$ is an orthonormal null basis ($k =$nullity $= 406$ in this run) and $f_\theta : \mathbb{R}^{178} \to \mathbb{R}^{k}$.

Architecture (`NullspaceNet`). Reuses `DirectODRegressor` as a trunk with **output dimension = nullity** (not 576):

```
x_scaled (178)
   → Linear 178→512 + GELU
   → Linear 512→256 + GELU
   → Linear 256→406                  (z coefficients)
         |
         ▼
ŷ = A⁺ x_raw  +  z @ Nᵀ             (hard particular + null correction)
         |
         ▼
softplus; zero diagonal; support mask
```

Training extras. Soft null penalty $\lambda_{\text{null}}\|A(N\mathbf{z})\|_2^2$ added to composite loss so that the learned correction stays approximately in $\ker(A)$ even after nonlinear constraints. After softplus / diagonal masking, exact null membership is **not** guaranteed (see null-space diagnostics).

Selected hyperparameters:

Hyperparameter	Value
hidden0 / hidden1	512 / 256
lr	0.001716044502975482
weight_decay	0.0007886714129990489
batch_size	64
forward_weight	1.0263495397682354
output_activation	softplus
null_penalty	0.029914693021302164
parameters (CSV)	327 318

What it measures. Learning a prior **restricted to** the null space. Best survey OD MAE in this run; middling synthetic OD and imperfect forward (soft penalty + post-activations).

Architecture and design comparison

Model	Predict uses A?	Learns OD prior?	Explicit null-space?	Constraint (reported)	Hypothesis class
moore_penrose	yes (A^+)	no	particular only	none	min-norm LS
tikhonov	yes	weak (ℓ_2)	no	none	Tikhonov LS
ridge	no	yes (linear)	implicit	none	linear $X \to Y$
physics_ridge	yes (refine)	yes + physics	implicit	none	Ridge + forward pull
pls	no	yes (latent linear)	no	none	PLS rank-k
mlp	loss only	yes (MLP)	no	relu	178→512→1024→576
residual_mlp	loss only	yes (ResBlocks)	no	softplus	178→512→Res×3→576
physics_residual_mlp	loss only	yes + survey base	no	relu	Res×1 + $\mathbf{y}_{\text{survey}}$
ae_*	loss only	yes (AE manifold)	no	softplus/relu	$X \to$ latent $\to$ OD
nullspace_mlp	yes (A^+, N)	yes (f_θ)	**yes**	softplus	$A^+x + Nz(x)$

Hyperparameter search and selection

Shared neural suggestions (`neural_train.suggest_neural_params`): `hidden0`∈{128,256,512}, `hidden1`∈{256,512}, `lr` log $[10^{-4}, 2\times10^{-3}]$, `weight_decay` log $[10^{-6}, 10^{-3}]$, `batch_size`∈{64,128,256}, `forward_weight`∈[0,1.5], `output_activation`∈{none,relu,softplus}.

Selection objective: validation **composite** (see Metrics), `n_trials`=8.

Validation HPO best values (`*/hpo_best.json` `best_value`):

Model	Val composite (HPO best)
physics_ridge	1807.137163
ridge	1807.155101
mlp	2229.268670
physics_residual_mlp	2345.978975
residual_mlp	2359.541547
ae_64_finetune	3277.275964
ae_128	3622.819995
pls	3626.762514
ae_64	3705.674493
ae_32	3776.059770
nullspace_mlp	4217.626729

Moore–Penrose: no HPO. Tikhonov HPO best_value: **Not available in the current experiment artifacts.**

Metrics

Implemented in `src/ml/rigorous_benchmark/metrics_suite.py`.

Forward (turning space):

$$\hat{\mathbf{x}} = \hat{\mathbf{Y}}_{\text{flat}}\, A^{\top}, \qquad \text{forward MAE/RMSE on } \mathbf{x} - \hat{\mathbf{x}}.$$

OD metrics: MAE, RMSE, R^2, Pearson, Spearman (capped `metrics_spearman_samples=500`), relative error, sparsity error, negative cells, diagonal violation.

Marginals: production / attraction MAE & RMSE.

Inverse-problem extras: `mean_l2_to_pinv`, `mean_rel_nullspace_leak_via_A`.

Composite (selection / ranking):

$$\text{score} = n\text{MAE}_{\text{OD}} + \alpha\, n\text{Fwd} + \beta\, n\text{Prod} + \gamma\, n\text{Attr},$$

with $\alpha = 1.0$, $\beta = \gamma = 0.5$ (`run_config.json`). Test composites use validation-set references (`selection.attach_composite`).

Unavailable in artifacts: entropy difference, trip-length / distance-bin errors, zone-distance stratified error.

Synthetic test results

Source: `outputs/benchmark/benchmark_results.csv` (exact floats). Tables are split so columns remain readable in Markdown and PDF.

OD accuracy (synthetic test)

Rank	Model	OD MAE	Pearson	Composite
1	mlp	646.575605	0.624761	1.852621
2	residual_mlp	657.214350	0.608259	1.965311
3	physics_residual_mlp	667.663681	0.586593	1.934136
4	nullspace_mlp	680.165591	0.494405	3.166864
5	ridge	697.985720	0.614995	1.506371
6	physics_ridge	698.036344	0.614942	1.506216
7	moore_penrose	731.185951	0.461912	3.390996
8	tikhonov	734.362355	0.429899	3.654341
9	ae_64_finetune	791.845282	0.357722	3.285658
10	pls	817.568364	0.301449	4.189618
11	ae_128	820.910334	0.274172	3.951162
12	ae_64	824.707534	0.253679	4.252426
13	ae_32	831.729739	0.235525	4.525160

Composite ranking (lower better): physics_ridge (1.506216) < ridge (1.506371) < mlp (1.852621) < …

Forward consistency and marginals (synthetic test)

Model	Forward RMSE	Prod MAE	Attr MAE
moore_penrose	8.536e-05	4705.106327	4706.964408
physics_ridge	0.002614	1100.299023	1093.898000
tikhonov	0.132546	5213.960843	5214.147098
ridge	0.247070	1100.213968	1093.815987
mlp	339.939142	1065.576747	1052.385210
physics_residual_mlp	344.249327	1159.121027	1158.029047
residual_mlp	372.933580	1173.803640	1165.446359
nullspace_mlp	798.842649	2783.060571	2802.462876
ae_64_finetune	1227.010749	1267.937464	1242.946557
ae_128	1618.408914	1572.240635	1546.554931
pls	1876.895504	1621.738799	1617.774040
ae_64	1906.380269	1487.594875	1485.121037
ae_32	2118.186015	1501.644228	1516.763000

Plots — synthetic test

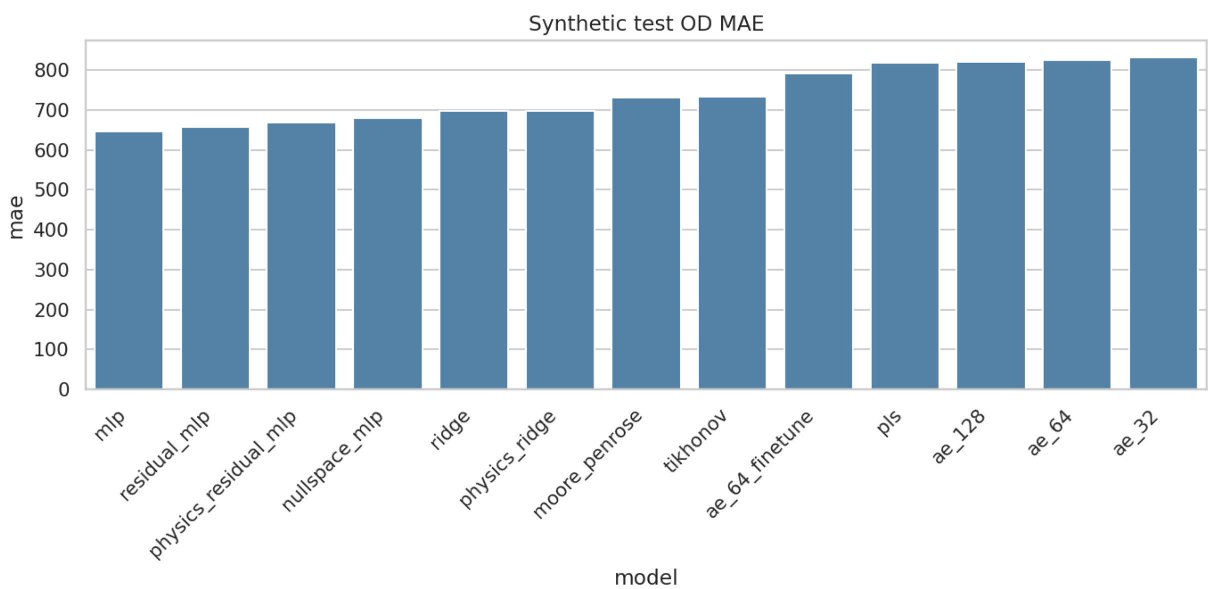

Figure 1: Synthetic test OD MAE

Figure — test OD MAE. Mean absolute error on the held-out 15 000 synthetic OD matrices (lower is better). Bars are ordered by model as produced by the figure script. The MLP bar is lowest (~647); residual / physics-residual / nullspace follow; AE and PLS form the high-error cluster (~792–832). This plot answers “who recovers the FP OD prior on in-distribution synthetics?” — not who is forward-consistent.

Figure — test forward RMSE. RMSE of $\hat{\mathbf{x}} = A\hat{\mathbf{y}}$ vs observed turns. Operator and ridge-family bars sit near zero (logically expected for methods that solve or refine with A). Neural / AE /

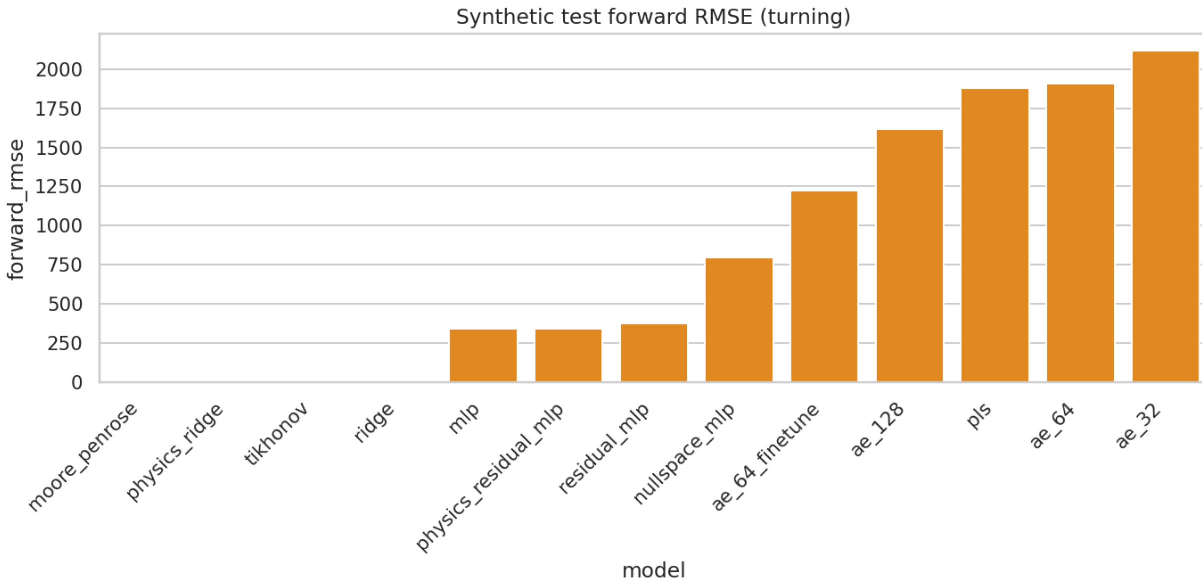

Figure 2: Synthetic test forward RMSE

PLS bars are orders of magnitude larger. Comparing this plot to OD MAE shows the classic trade-off: low OD error does not imply low forward error when the null space is large.

Figure — OD vs forward Pareto (test). Each point is one model. Ideal is bottom-left (low OD MAE and low forward RMSE). Ridge / physics_ridge occupy the low-forward, mid-OD region; MLP occupies lower OD but higher forward; AE/PLS sit upper-right (worse on both axes under this protocol). There is no model in the ideal corner — evidence that a single scalar ranking is insufficient.

Figure — test Pearson correlation. Cell-wise correlation between predicted and true OD (higher better). Tracks OD MAE ordering roughly: MLP / residual / ridge highest (~0.61–0.62); AE_32 lowest (~0.24). Pearson rewards linear association even when absolute scale errs; still, AE/PLS remain weak.

Figure — test production MAE. Error on row sums (trips leaving each origin). Moore–Penrose and Tikhonov are catastrophic (~4700–5200) despite near-perfect forward — the particular / Tikhonov solutions do not match FP total productions. Supervised models keep production MAE near ~1100–1600; nullspace_mlp is worse (~2783), consistent with unconstrained total demand along null directions after softplus.

Figure — test attraction MAE. Column-sum errors. Pattern mirrors production MAE: operator methods fail on marginals; supervised models are far better; nullspace sits in between.

Interpretation of synthetic rankings

- **MLP wins OD MAE** by learning a strong synthetic prior in the large null space, at the cost of non-zero forward error (~340 RMSE).
- **Physics Ridge / Ridge win composite** because forward error is essentially zero while OD MAE remains competitive (~698); composite heavily weights forward/marginals after val normalization.
- **Moore–Penrose / Tikhonov** nearly perfect forward consistency but poor OD MAE and disastrous production/attraction MAE — they recover a particular solution, not the FP prior.

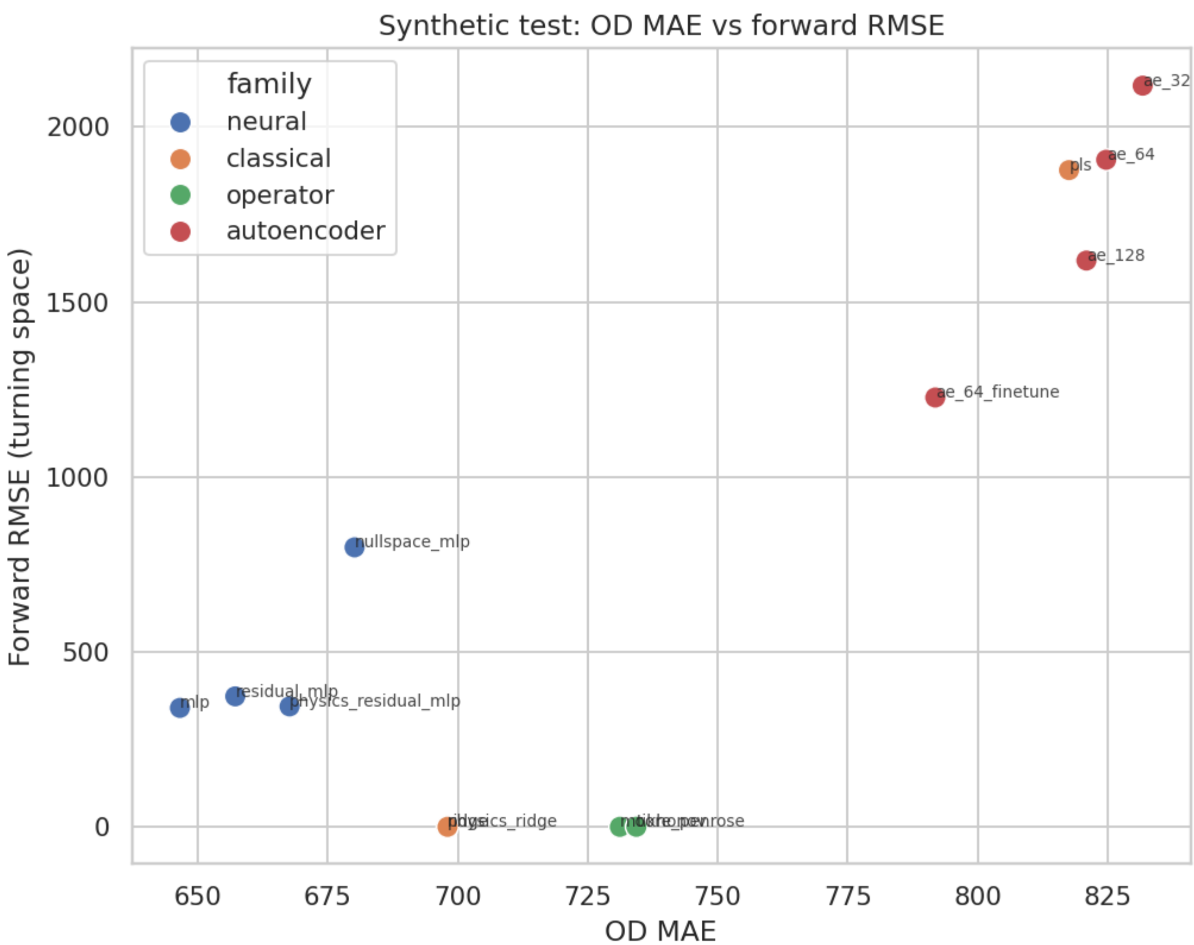

Figure 3: OD MAE vs forward RMSE (test)

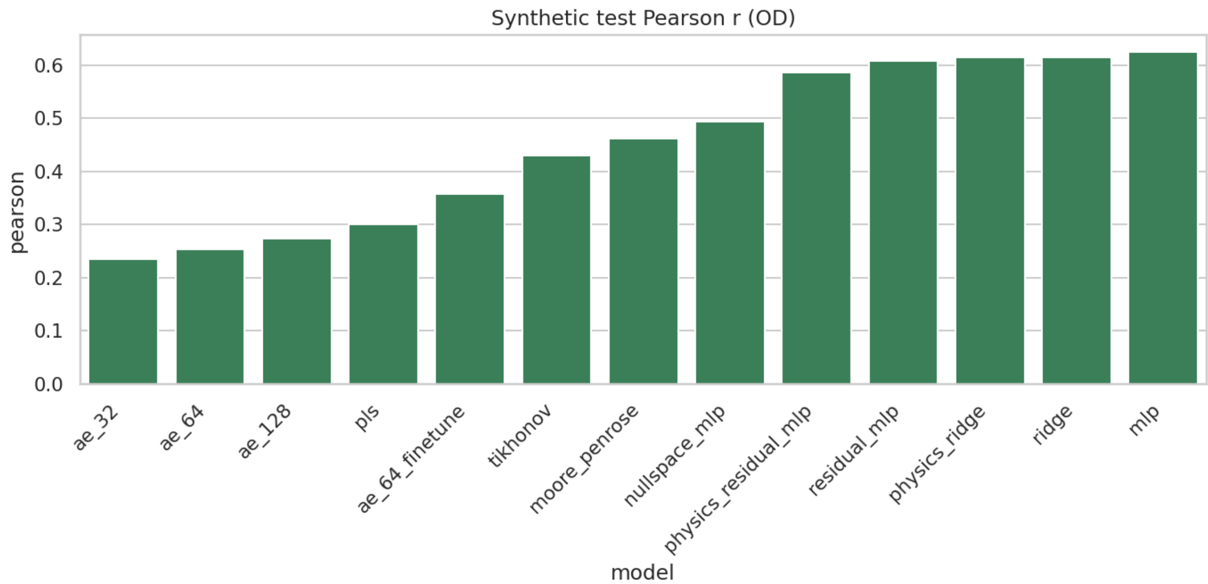

Figure 4: Pearson

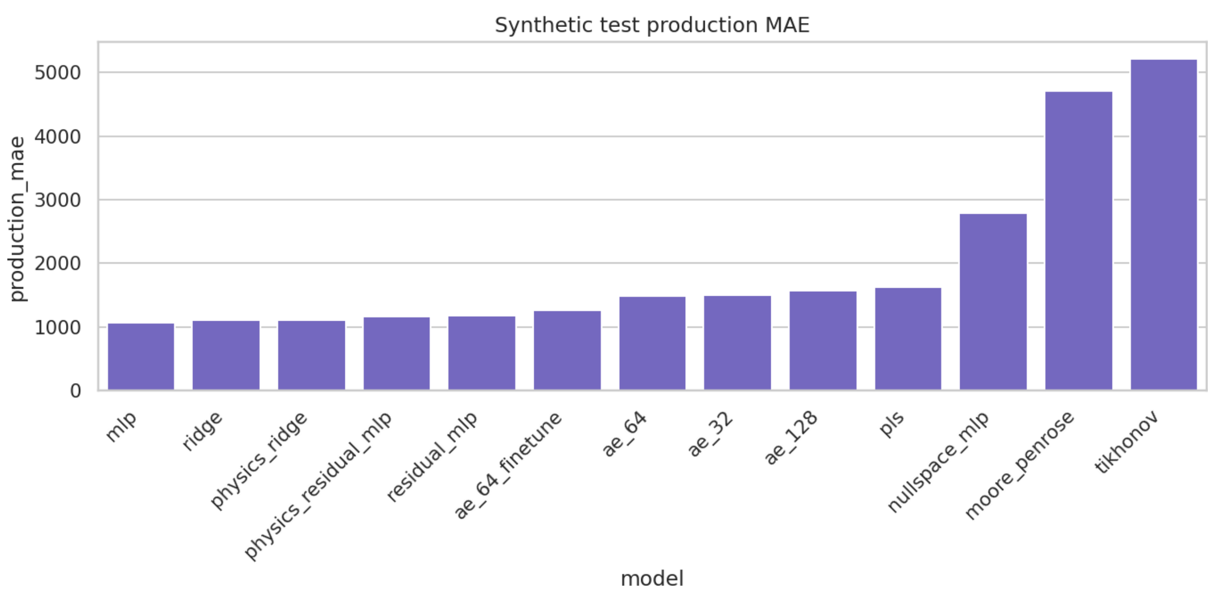

Figure 5: Production MAE

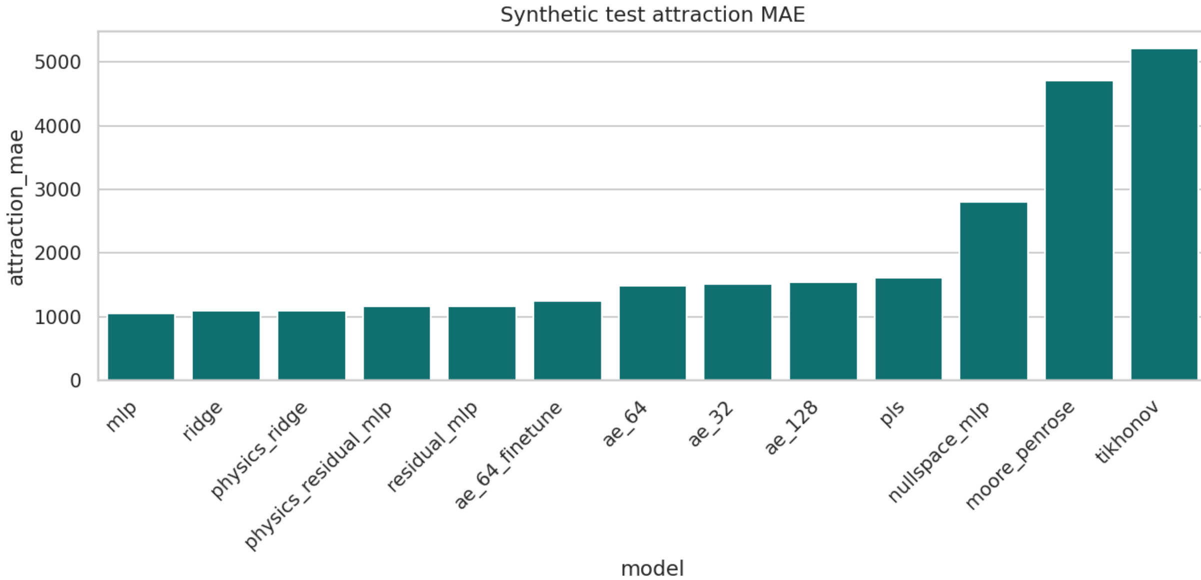

Figure 6: Attraction MAE

- **Nullspace MLP** sits between: better OD than pinv, worse forward than ridge, weaker synthetic OD than plain MLP — consistent with a constrained hypothesis class $A^{+}\mathbf{x} + N\mathbf{z}$.
- **AE / PLS** underperform on both OD and forward under this HPO budget (`n_trials`=8, capped train).

Survey inference results

Source: `outputs/benchmark/survey_inference/survey_results.json` (one survey OD realization). Tables split for readability.

Survey OD accuracy

Rank	Model	OD MAE	Pearson
1	nullspace_mlp	395.114834	0.488878
2	pls	418.039700	0.374540
3	moore_penrose	444.140140	0.418043
4	ae_64_finetune	451.962092	0.329399
5	ae_32	465.063583	0.197415
6	tikhonov	466.588330	0.243907
7	ae_64	485.612941	0.236571
8	physics_residual_mlp	512.339152	0.211974
9	ae_128	514.507728	0.222108
10	mlp	527.377343	0.211658
11	residual_mlp	527.626517	0.219909
12	ridge	664.032835	0.168075
13	physics_ridge	664.149324	0.168054

Survey forward consistency, marginals, and null-space proximity

Model	Forward RMSE	Prod MAE	Attr MAE	ℓ_2 to pinv	Rel. null leak
moore_penrose	3.607e-05	6046.130955	6046.130955	0.0	0.0
physics_ridge	0.005382	5991.510249	6050.212830	23615.960148	1.26e-06
ridge	0.156431	5991.069385	6049.726024	23615.524453	3.66e-05
tikhonov	0.259294	7242.966101	7242.966101	7611.792835	6.07e-05
physics_residual_mlp	280.397207	6114.335692	6003.717958	23056.313305	0.065592
mlp	325.749433	5931.444914	5806.048229	24038.045183	0.076201
nullspace_mlp	374.365287	4343.379282	4427.213110	11648.343245	0.087573
residual_mlp	411.646022	5791.312941	5981.157951	22978.867064	0.096294
ae_64_finetune	971.987312	6034.791032	6335.327453	18963.821577	0.227372
ae_128	1255.887706	6131.061538	6011.479929	19773.213635	0.293784
pls	1292.271232	6314.272479	6298.194063	16653.608115	0.302295
ae_64	1586.096551	5900.370275	6212.968683	18560.692864	0.371028
ae_32	1867.307089	6187.208738	6351.626751	17293.434131	0.436810

Survey R^2: recorded as `NaN` in JSON for all models (degenerate total sum of squares) — **Not available as a meaningful scalar in the current experiment artifacts.**

Plots — survey

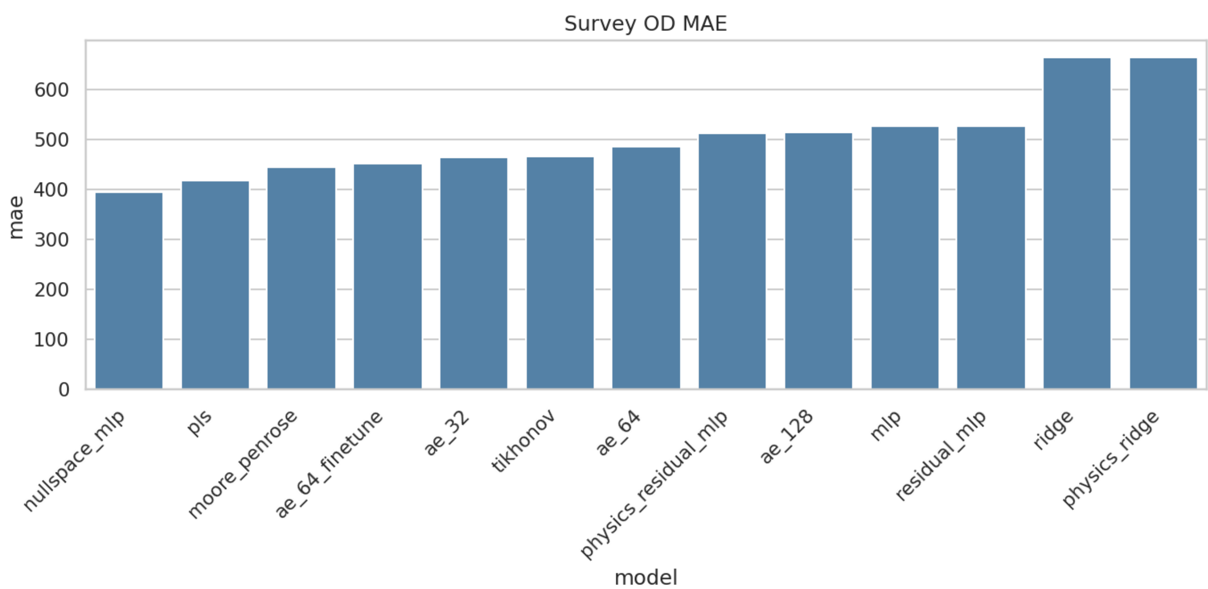

Figure 7: Survey OD MAE

Figure — survey OD MAE. Absolute error vs the single held-out survey OD matrix. Ranking **reverses** relative to synthetic test: `nullspace_mlp` is best (~395), then PLS / pinv; MLP falls to ~527; ridge / physics_ridge are worst (~664). Read this as out-of-distribution transfer under fixed A, not as a second synthetic leaderboard.

Figure — survey forward RMSE. Operator and ridge-family methods remain near zero. Neural models keep moderate forward error; AE/PLS again large. Forward excellence on survey does **not** predict OD MAE (ridge has ~0 forward and worst OD).

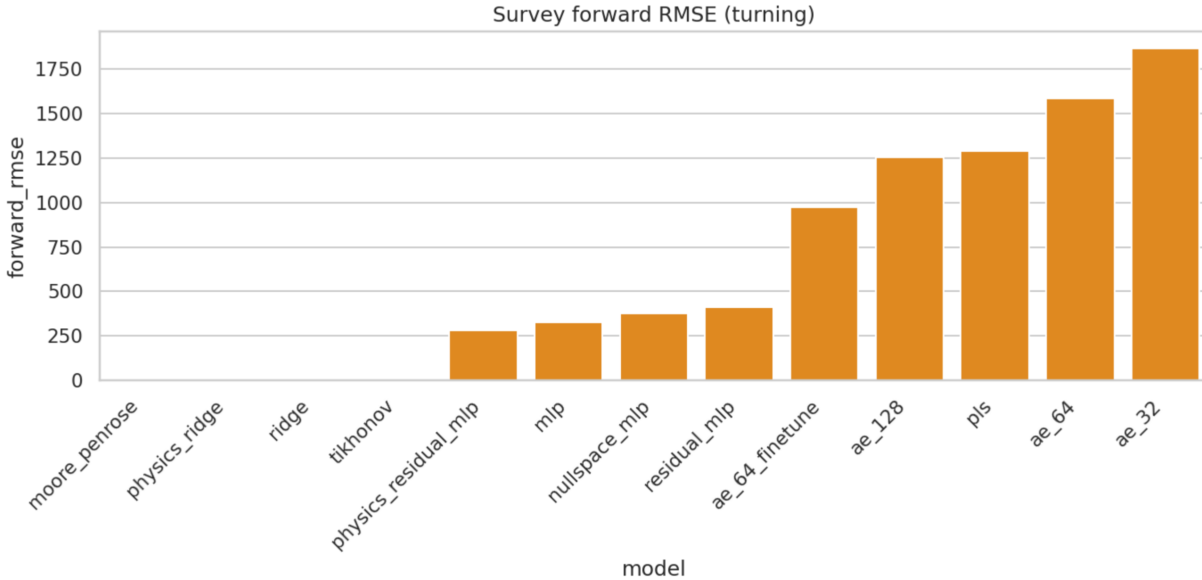

Figure 8: Survey forward RMSE

Figure — survey OD vs forward scatter. Nullspace_mlp sits at comparatively low OD MAE with moderate forward error. Moore–Penrose is near-zero forward with mid OD MAE. Ridge/physics_ridge hug the forward axis but sit far right on OD MAE. The spread visualizes why "best model" depends on the chosen axis.

Disagreement with synthetic test

MLP best on synthetic OD; **nullspace_mlp** best on survey OD; ridge/physics_ridge best on synthetic composite but **worst** survey OD MAE. Explained by distribution shift (scale diagnostics above) and nullity: models that memorize the FP prior transfer poorly; models that stay near the affine solution set $A^+\mathbf{x} + N\mathbf{z}$ or the particular solution transfer better on this single survey matrix.

Null-space diagnostics

Artifact-derived (`docs/figures/od_benchmark/nullspace_diagnostics.json`), using survey `moore_penrose_od.npy` vs `nullspace_mlp_od.npy` and `operator/a_pinv.npy` (no checkpoint restore):

Quantity	Value
Survey OD MAE (pinv)	444.140139
Survey OD MAE (nullspace_mlp)	395.114833
$\|\hat{\mathbf{y}}_{\text{ns}} - A^+\mathbf{x}\|$	11685.562730
$\|\hat{\mathbf{y}}_{\text{ns}} - \hat{\mathbf{y}}_{\text{pinv}}\|$	11648.343245
$\|A(\hat{\mathbf{y}}_{\text{ns}} - \hat{\mathbf{y}}_{\text{pinv}})\|$	4994.655938
Relative leak via A	0.087573
Mean	cell

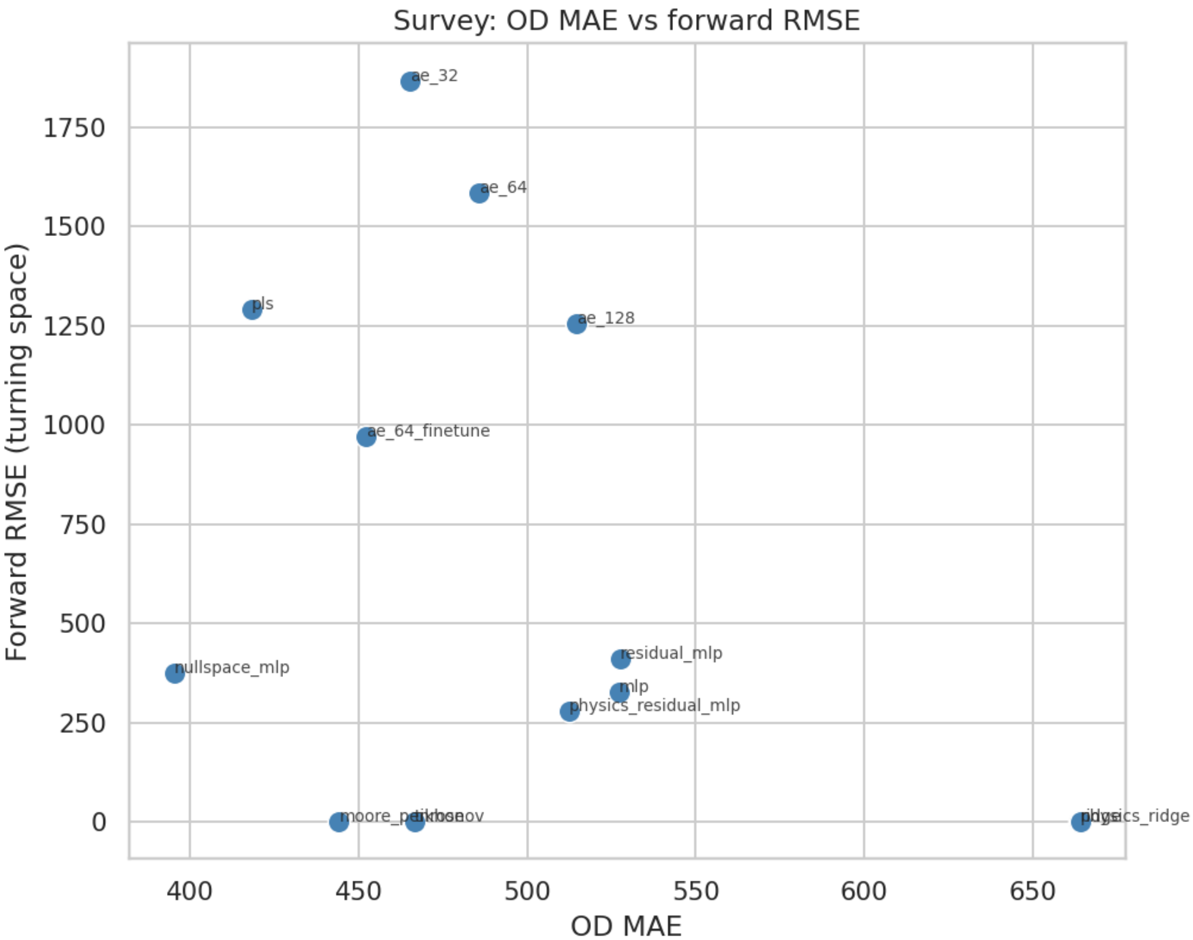

Figure 9: Survey OD vs forward

Interpretation (hypothesis language): nullspace_mlp applies a large OD-space correction relative to $A^{+}\mathbf{x}$; residual forward leakage (~0.088 relative) shows the soft null penalty and output constraints do not enforce $AN\mathbf{z} = \mathbf{0}$ exactly after softplus/masking.

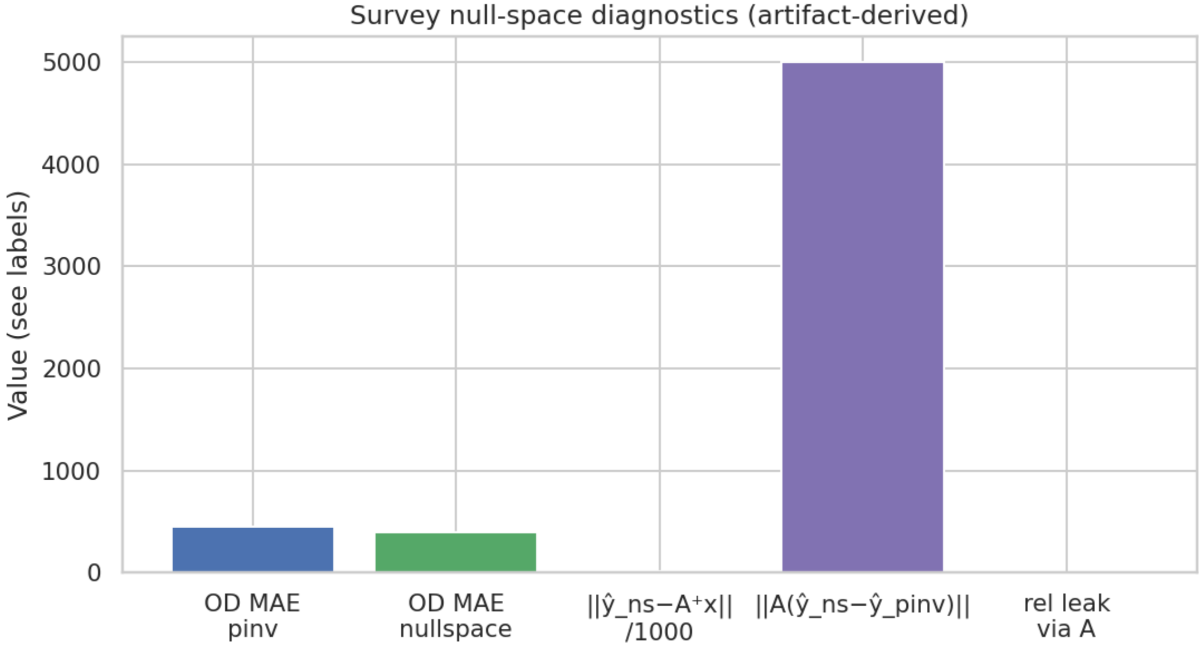

Figure 10: Nullspace diagnostics

Figure — nullspace diagnostics bars. Summarizes the magnitudes above: OD improvement vs pinv, size of the correction $\|y_{\text{ns}} - y_{\text{pinv}}\|$, and residual $\|A\Delta y\|$. Large correction with non-zero leak is the expected signature of a soft-constrained null-space learner.

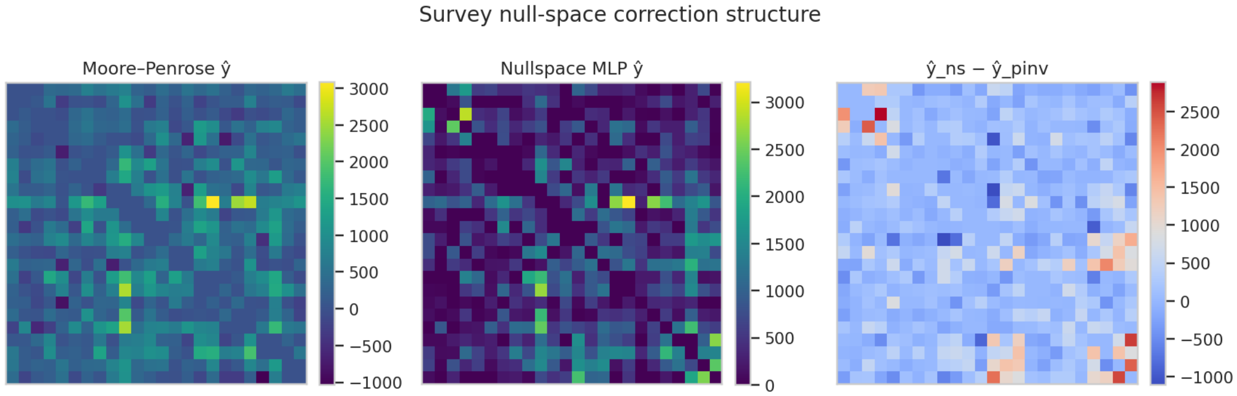

Figure 11: Nullspace correction heatmaps

Figure — nullspace correction heatmaps. Spatial structure of $y_{\text{ns}} - y_{\text{pinv}}$ (and related panels from the script). Bright corridors are OD pairs where the MLP injects null-space demand that pinv cannot express; this is the learned prior component.

Failure modes

Failure mode	Evidence	Models implicated
Perfect forward, wrong OD	Forward RMSE ≈ 0, high OD/prod MAE	moore_penrose, tikhonov, ridge, *physics_ridge*
Good synthetic OD, weak survey OD	MLP #1 test / #10 survey	mlp, residual_mlp
Forward collapse	Forward RMSE ≫ 10^3	ae_*, pls
Negatives	`negative_cells` > 0 on test	ridge, physics_ridge, moore_penrose, tikhonov, pls
Marginal mismatch	Prod/attr MAE ≫ OD MAE	pinv, tikhonov, nullspace (survey total demand error 55607.5)

*Ridge family: near-perfect forward **and** competitive synthetic OD, but survey OD collapses — prior mismatch, not forward failure.

Heatmap analysis

Survey heatmaps (primary)

Each figure shows true survey OD, model prediction, and absolute error as 24×24 panels (same color scale within a figure as produced by the script).

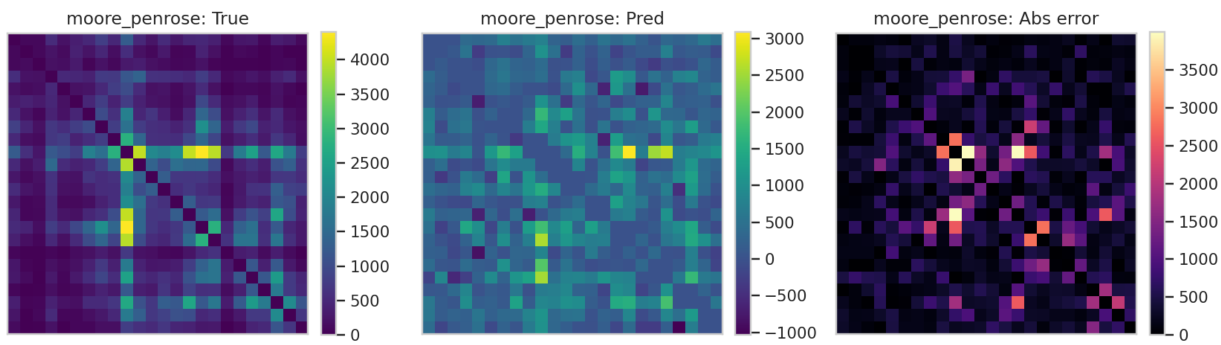

Figure 12: moore_penrose

Survey — Moore–Penrose. Prediction is the min-norm particular solution. Abs-error highlights OD cells where the survey prior disagrees with A^+x. Forward is essentially exact; residual structure is almost pure null-space disagreement.

Survey — Tikhonov. Visually close to pinv at the selected tiny λ; slightly damped high-frequency content. Error map remains large in absolute OD units despite near-zero forward RMSE.

Survey — Physics Residual MLP. Anchored to survey base OD during training, yet still shows structured abs-error: the network learns a correction that does not fully recover the survey matrix under the synthetic training distribution / forward pressure.

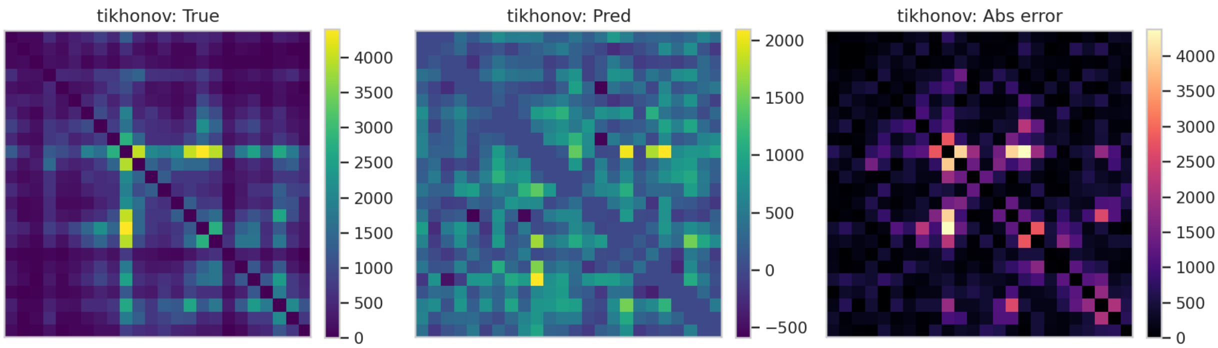

Figure 13: tikhonov

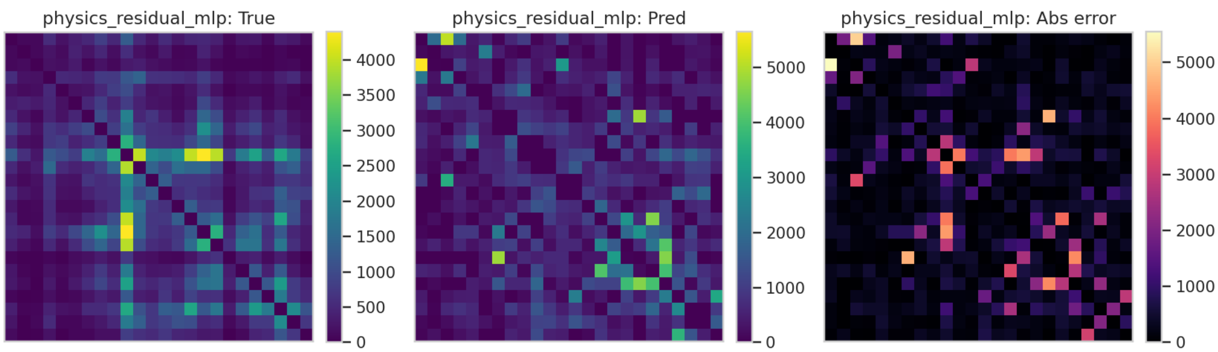

Figure 14: physics_residual_mlp

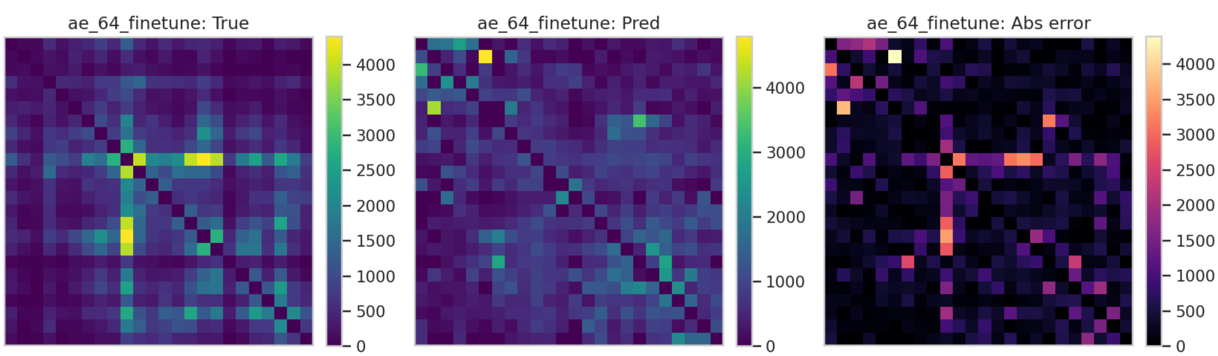

Figure 15: ae_64_finetune

Survey — AE-64 finetune. Predictions are smoother / more compressed (latent bottleneck). Abs-error is broad; forward RMSE is large, consistent with decoder drift off the A-consistent set.

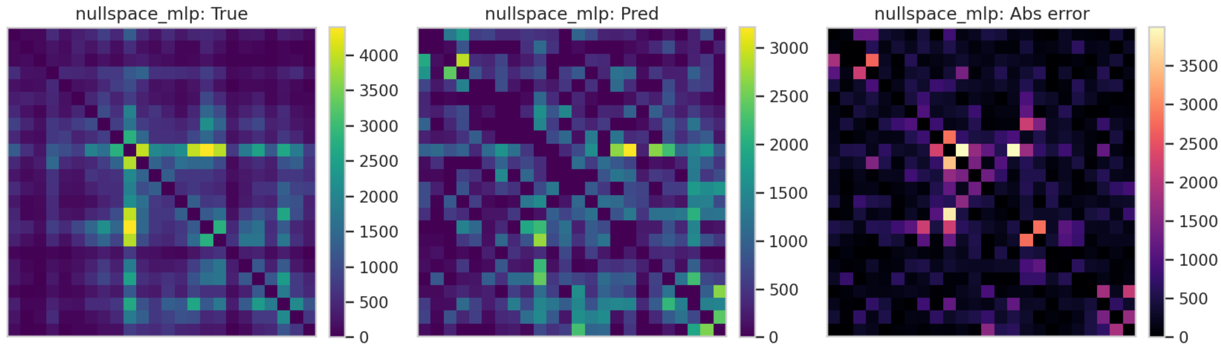

Figure 16: nullspace_mlp

Survey — Nullspace MLP. Closest overall OD match among the five visualized models (MAE 395). Abs-error is reduced on many corridors relative to pinv while retaining more structure than AE; remaining errors concentrate where softplus / soft null penalty limit exact affine recovery.

Synthetic sample heatmaps

Fixed first test index (`test_idx[0]=0` in `split_indices.npz`; local row 0 of each `test_pred.npy`). Meta: `figures/od_benchmark/synthetic_heatmap_meta.json`.

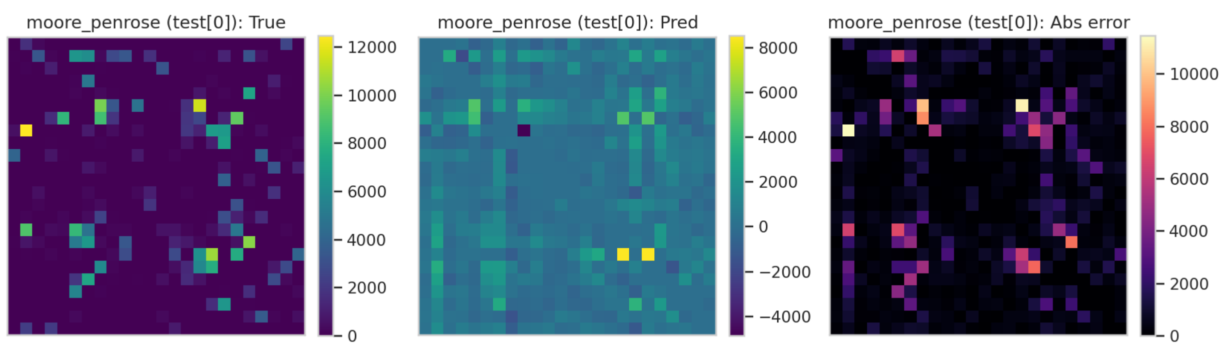

Figure 17: synth moore_penrose

Synthetic sample — Moore–Penrose. Particular solution vs one FP draw. Abs-error shows where FP mass lives off the min-norm solution.

Synthetic sample — Tikhonov. Nearly identical story to pinv for this λ.

Synthetic sample — Physics Residual MLP. Closer to the FP sample than operator methods on many cells, but still imperfect; survey-base residual is a mismatched anchor for pure synthetics.

Synthetic sample — AE-64 finetune. Under-resolved corridors and larger abs-error than residual MLP family on this index.

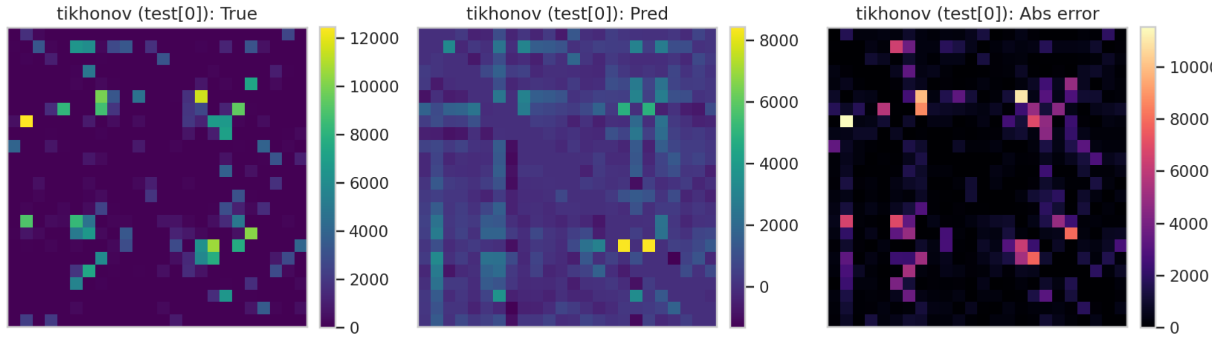

Figure 18: synth tikhonov

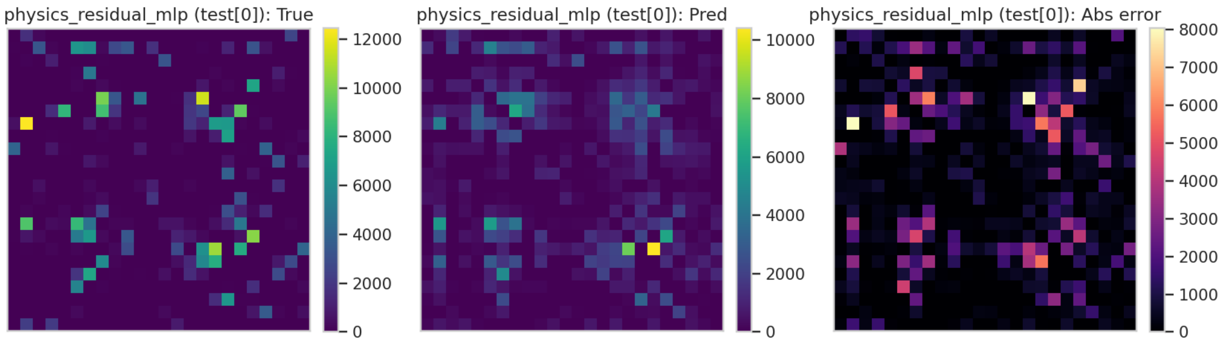

Figure 19: synth physics_residual_mlp

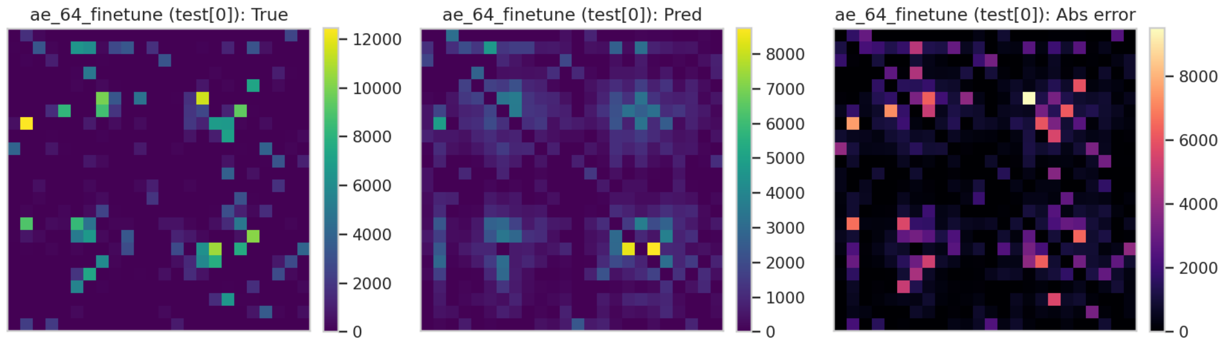

Figure 20: synth ae_64_finetune

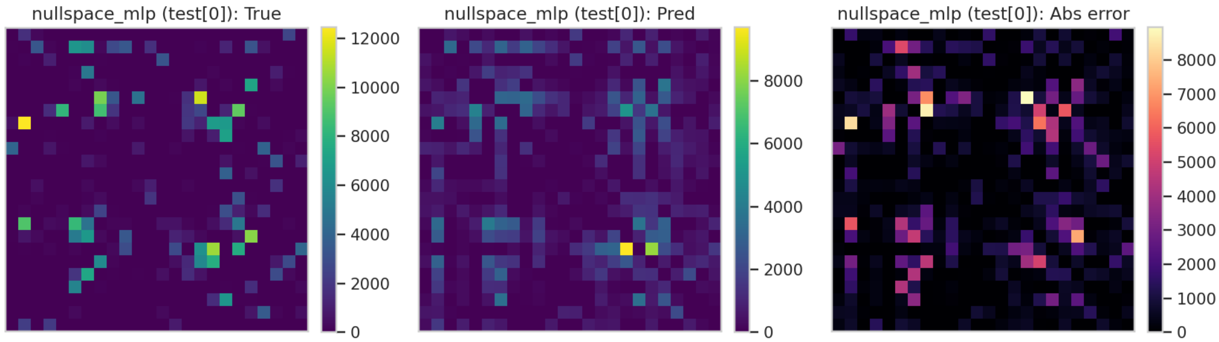

Figure 21: synth nullspace_mlp

Synthetic sample — Nullspace MLP. Improves on pinv for this FP sample by adding a null correction; residual abs-error remains visible on heavy corridors.

Trip-distance / bin error heatmaps: **Not available in the current experiment artifacts.**

Error structure notes

From survey metrics:

- Support Jaccard highest for PLS (1.0) and AE variants (~0.87–0.95); operator methods ~0.77–0.78.
- Pred sparsity often ≫ true sparsity (true ≈ 0.0417): models over-sparsify.
- Diagonal violation ≈ 0 after constraints; raw pinv has tiny diagonal leak (8.4×10^{-8}).

Cell-wise spatial Moran's I / corridor-wise errors: **Not available in the current experiment artifacts.**

Computational profile

From `benchmark_results.csv` / JSON (seconds). Missing entries marked unavailable.

Model	Parameters	Train time (s)	Inference time (s)
moore_penrose	N/A	Not available	Not available
tikhonov	N/A	Not available	Not available
ridge	N/A	Not available	Not available
physics_ridge	N/A	Not available	Not available
pls	N/A	Not available	Not available
mlp	N/A	Not available	Not available
residual_mlp	1 963 072	44.466550	1.201899
physics_residual_mlp	912 448	57.363631	1.143143
ae_32	556 000	99.455975	1.032126
ae_64	576 512	97.370932	1.036879
ae_128	617 536	108.651230	0.942379
ae_64_finetune	576 512	128.678197	1.098481
nullspace_mlp	327 318	23.215592	1.081002

Full benchmark wall time: 628.862 s (`run_summary.json`). Peak GPU memory: **Not available in the current experiment artifacts.**

Multi-criteria summary

Criterion	Winner (artifacts)	Value
Synthetic OD MAE	mlp	646.575605
Synthetic composite	physics_ridge	1.506216
Synthetic forward RMSE	moore_penrose	8.536×10^{-5}
Survey OD MAE	nullspace_mlp	395.114834
Survey forward RMSE	moore_penrose	3.607×10^{-5}
Survey Pearson	nullspace_mlp	0.488878

Rankings by family

Family	Best synthetic OD	Best survey OD	Best synthetic composite
operator	moore_penrose (731.19)	moore_penrose (444.14)	moore_penrose (3.391)
classical	ridge (697.99)	pls (418.04)	physics_ridge (1.506)
neural	mlp (646.58)	nullspace_mlp (395.11)	mlp (1.853)
autoencoder	ae_64_finetune (791.85)	ae_64_finetune (451.96)	ae_64_finetune (3.286)

Constraint ablations

Source: `outputs/benchmark/ablations/*_ablation.json`. Strategies: `none`, `relu`, `softplus`, `clip`, `clip_ipf`. Only mlp / ridge / moore_penrose were ablated.

mlp (baseline already relu-constrained)

Strategy	OD MAE	Forward MAE	Prod MAE	Attr MAE
none	646.575605	227.761460	1065.576750	1052.385208
relu	646.575605	227.761460	1065.576750	1052.385208
softplus	646.584119	227.911837	1065.824897	1052.614491
clip	646.575605	227.761460	1065.576750	1052.385208
clip_ipf	642.588341	338.328244	0.001530	0.000857

ridge

Strategy	OD MAE	Forward MAE	Prod MAE	Attr MAE
none	697.985720	0.139894	1100.213968	1093.815987

Strategy	OD MAE	Forward MAE	Prod MAE	Attr MAE
relu	652.740765	306.835866	1499.851585	1495.557018
softplus	652.743142	306.876982	1499.897464	1495.603761
clip	652.740765	306.835866	1499.851585	1495.557018
clip_ipf	637.722442	384.345463	0.001644	0.000856

moore_penrose

Strategy	OD MAE	Forward MAE	Prod MAE	Attr MAE
none	731.185951	3.738e-05	4705.106327	4706.964408
relu	666.719798	420.051229	3504.947987	3501.099084
softplus	666.696159	420.112940	3503.676531	3499.829185
clip	666.719798	420.051229	3504.947987	3501.099084
clip_ipf	729.222855	1450.579031	0.001985	0.000857

Takeaway: Nonnegativity (`relu`/`clip`) improves OD MAE for unconstrained classical/operator preds but **destroys** near-zero forward error. `clip_ipf` forces near-perfect marginals and worsens forward consistency — do not treat IPF as a free OD MAE win without reporting forward.

Ablations for other models: **Not available in the current experiment artifacts.**

Theory ↔ implementation map

Theory	Implementation path
$\mathbf{x} = A\mathbf{y}$	`data.verify_forward_map`, `metrics_suite.forward_predictions`
A^+	`operator.compute_operator`, `pinv_predict`
Tikhonov	`models/tikhonov.py`
Supervised Ridge	`models/ridge.py`
Physics refine	`models/physics_ridge.physics_refine`
$\hat{\mathbf{y}} = A^+\mathbf{x} + Nf(\mathbf{x})$	`models/nullspace_mlp.py`
Survey base residual	`data.base_od_flat` + `neural_train.logits_to_od(..., residual=True)`
Composite selection	`metrics_suite.composite_score`, `selection.attach_composite`
Constraints	`constraints.apply_constraint_strategy`
Benchmark driver	`src/ml/rigorous_benchmark/run.py`

Reproduction

```
# Full benchmark (long; uses paths in BenchmarkConfig / CLI)
python -m src.ml.rigorous_benchmark.run

# Smoke
python -m src.ml.rigorous_benchmark.run --smoke
```

```
# Regenerate documentation figures from existing artifacts
python scripts/generate_od_benchmark_doc_figures.py
```

Environment note: project targets conda env `ODEstimation` (see `requirements.txt`).

Limitations

1. **One survey realization** — survey rankings are a single matrix (`EstimatedODMatrix.npy`), not a survey ensemble.
2. **Fixed A** — no stochastic assignment / congestion feedback.
3. **Synthetic → survey shift** — confirmed by scale diagnostics; FP prior ≠ survey prior.
4. **HPO budget** — 8 trials, 5k HPO train subsample, 25k final train cap: neural rankings are under a memory-safe protocol, not an exhaustive search.
5. **Missing timings/params** for early classical/MLP rows in the results table.
6. **No trip-length / entropy metrics** in saved artifacts.
7. SVD rank in this dump (170) differs from older analysis text (174); both are reported, not conflated.

Next steps

Motivated by the nullity (406) and spatial OD structure: graph neural estimators on the zone graph or bipartite OD graph could encode corridor structure more explicitly than flat MLPs, while retaining a null-space head or forward loss. This is a **motivation**, not a claim of superiority — no GNN results exist in `outputs/benchmark/`.

Other next steps: multi-survey evaluation, equilibrium-consistent $A(\mathbf{y})$, richer HPO, and reporting train-time for all models.

References

Citations verified against existing `docs/*.tex` bibliographies (Ortúzar & Willumsen; Cascetta; Furness; Deming & Stephan; Wilson). Classical inverse-problem / Tikhonov / Moore–Penrose statements follow the implementations and standard linear-algebra definitions used in code comments; standalone Tikhonov (1963) / Penrose (1955) bibliographic entries are **not** present in those `.tex` files and are therefore not fabricated here as numbered bibitems.

1. J. de D. Ortúzar and L. G. Willumsen, *Modelling Transport*, 4th ed., Wiley, 2011. (`ortuzar2011` in `docs/fp_od_generation_methodology.tex`, `docs/synthetic_od_method.tex`, `docs/od_estimation_methodology.tex`)
2. E. Cascetta, "Estimation of trip matrices from traffic counts and survey data," *Transportation Research Part B*, 1984. (`cascetta1984`)
3. R. Furness, "Trip forecasting," *Traffic Engineering and Control*, 1962. (`furness1962`)
4. W. E. Deming and F. F. Stephan, "On a least squares adjustment of a sampled frequency table…," *Annals of Mathematical Statistics*, 1940. (`deming1940`)
5. A. G. Wilson, *Entropy in Urban and Regional Modelling*, Pion, 1970. (`wilson1970` in FP methodology)

Artifact index and PDF

Path	Role
`outputs/benchmark/benchmark_results.csv`	Test leaderboard
`outputs/benchmark/benchmark_results.json`	Full metrics
`outputs/benchmark/survey_inference/survey_results.json`	Survey metrics
`outputs/benchmark/*/best_params.json`	Fit hyperparameters
`outputs/benchmark/operator/operator_meta.json`	SVD summary
`outputs/benchmark/data/leakage_report.json`	Split / leakage
`outputs/benchmark/run_config.json`	Caps / weights / paths
`outputs/benchmark/ablations/`	Constraint ablations
`docs/figures/od_benchmark/`	Doc figures + `nullspace_diagnostics.json`

PDF render (pandoc + XeLaTeX):

```
pandoc docs/OD_INVERSE_ESTIMATION_MODEL_BENCHMARK.md \
  -o docs/OD_INVERSE_ESTIMATION_MODEL_BENCHMARK.pdf \
  --from markdown --resource-path=docs --pdf-engine=xelatex \
  -V geometry:margin=1in -V mainfont="DejaVu Sans" -V monofont="DejaVu Sans Mono"
```

Markdown remains the editable source of truth; mermaid diagrams appear as fenced code in the PDF unless a mermaid filter is added.